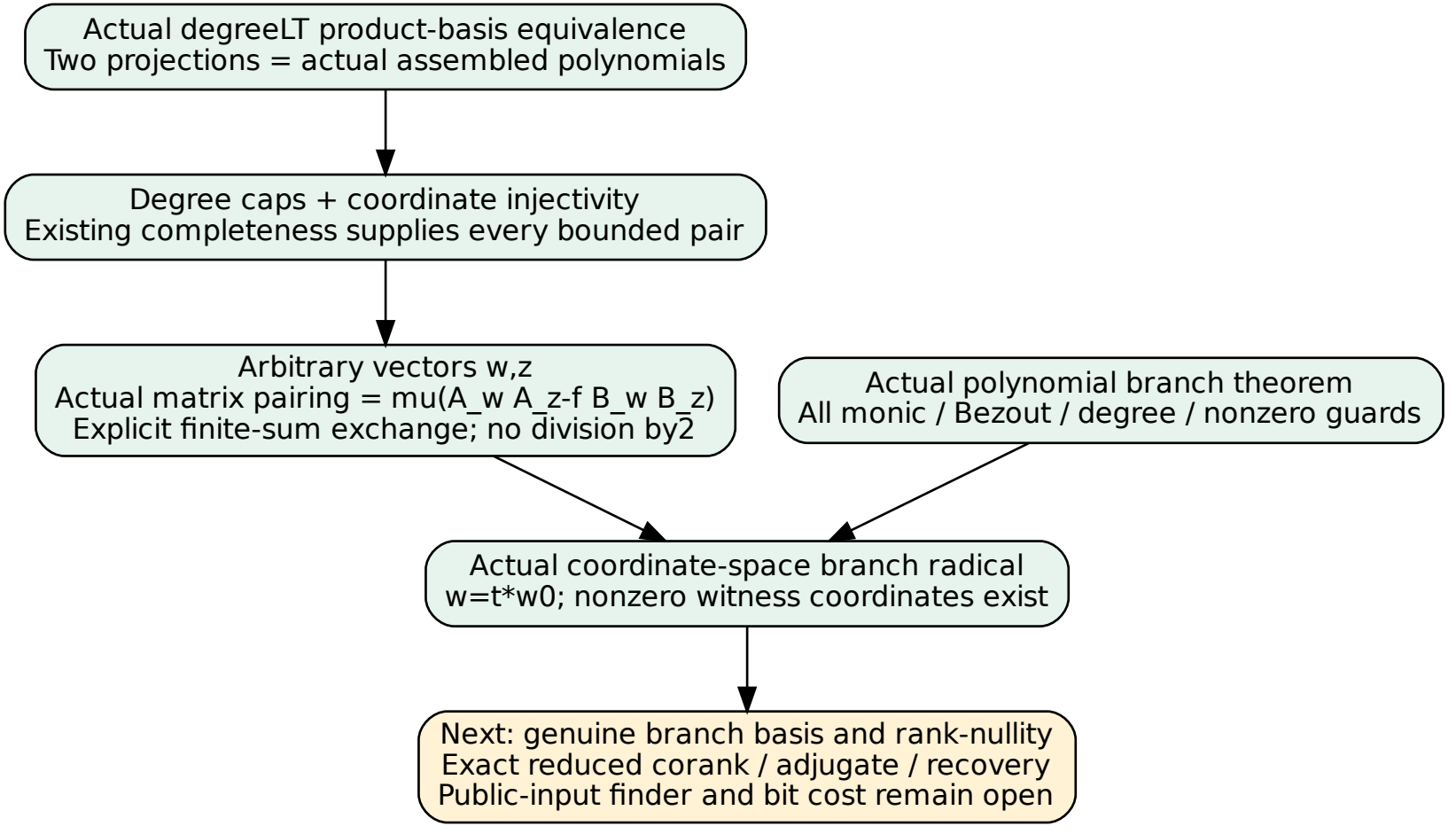
Actual degreeLT product-basis equivalence
Two projections = actual assembled polynomials
Degree caps + coordinate injectivity
Existing completeness supplies every bounded pair
Arbitrary vectors w,z
Actual matrix pairing = mu(A_w A_z-f B_w B_z)
Explicit finite-sum exchange; no division by2
Actual polynomial branch theorem
All monic / Bezout / degree / nonzero guards
Actual coordinate-space branch radical
w=t*w0; nonzero witness coordinates exist
Next: genuine branch basis and rank-nullity
Exact reduced corank / adjugate / recovery
Public-input finder and bit cost remain open